EPFL - Computer Science

CS-401 Applied Data Analysis

Lecture 05 — October 10, 2025

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Regression for disentangling data

Linear Regression

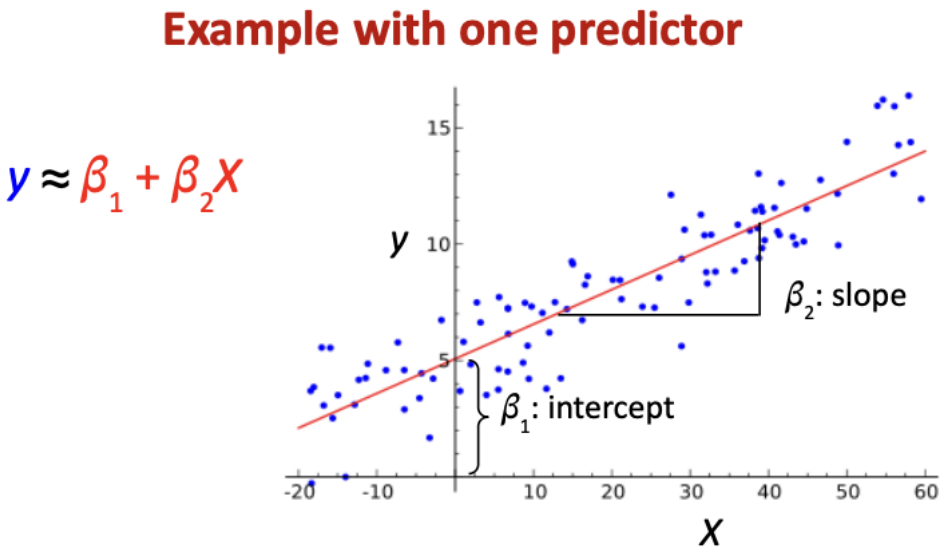
- Given: n data points (X, y). where X_i is k-dimensional vector of predictors (a.k.a. Features) of i-th data point, and y_i is scalar outcome
- Goal: find the optimal coefficient vector $\beta = (\beta_1, ..., \beta_k)$ for approximating the y_i's as linear function of the X_i's

$$y_i = X_i\beta + \epsilon_i$$

$$= \beta_I X_{i1} + ... + \beta_k X_{ik} + \epsilon_i. \text{ , for i = 1, ..., n}$$

where ϵ_i are the error terms that should be as small as possible

- X_n usually the constant 1 (by def) $\Rightarrow$ β_1 a constant intercept



Optimality criterion: least squares

$$y_i = X_i\beta + \epsilon_i, \text{ for i} = 1, ..., \text{n}$$

- Intuitively, want errors ϵ_i to be as small as possible
- Technically, want sum of squared errors as small as possible $\Leftrightarrow$ find $\hat{\beta}$ such that we minimize

$$\sum_{i=1}^{n} (y_i - X_i\hat{\beta})^2.$$

Use cases of regression

- Prediction: use fitted model to estimate outcome y for a new X not seen during model fitting
- Descriptive data analysis: compare average outcomes across subgroups of data (today!)
- Casual modeling: understand how outcome y changes when you manipulate predictors X (next lecture is about causality, although not primarily using regression)

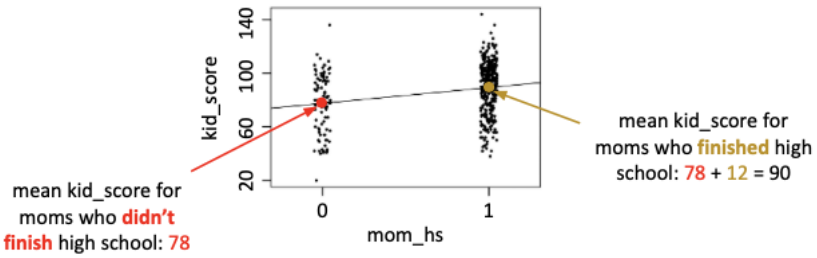
Regression as comparison of average outcomes

Example with one binary predictor X_i

- X_i = `mom_hs` = "Did mother finish high school?" $\in 0, 1$ where 0 = No, and 1 = Yes
- y_i = `kid_score` = child's score on cognitive test $\in [0, 140]$

$$y_i = \beta_i + \beta_2 X_i + \epsilon_i.$$

$$\texttt{kid_score} = 78 + 12 \cdot \texttt{mom_hs} + \text{error}$$



- Intercept β_1: mean outcome for data points i with $X_i = 0$
- Slope β_2: difference in mean outcomes between data points with $X_i = 1$ and data points with $X_i = 0$
- **Reason**: means minimize least-squares criterion:

$$\sum_{i=1}^{n} y_i - m^2 \text{ is minimized w.r.t. m when}$$
$$-2\sum_{i=1}^{n} (y_i - m) = 0, \text{ i.e., when m} = (1/n)\sum_{i=1}^{n} y_i$$

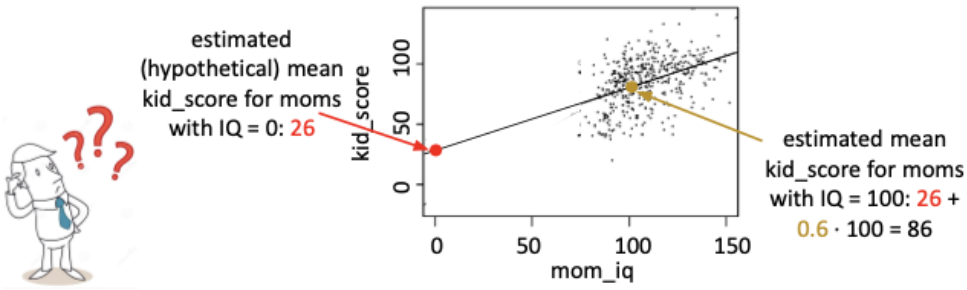
Mean monkey

So why not just compute the two means separately and them compare then?

Example with one continuous predictor X_i

- X_i = `mon_iq` = mother's IQ score $\in [70, 140]$
- y_i = `kid_score` = child's score on cognitive test $\in [0, 140]$

$$y_i = \beta_1 + \beta_2 X_i + \epsilon_i$$
$$\texttt{kid_score} = 26 + 0.6 \cdot \texttt{mon_iq} + \text{error}$$

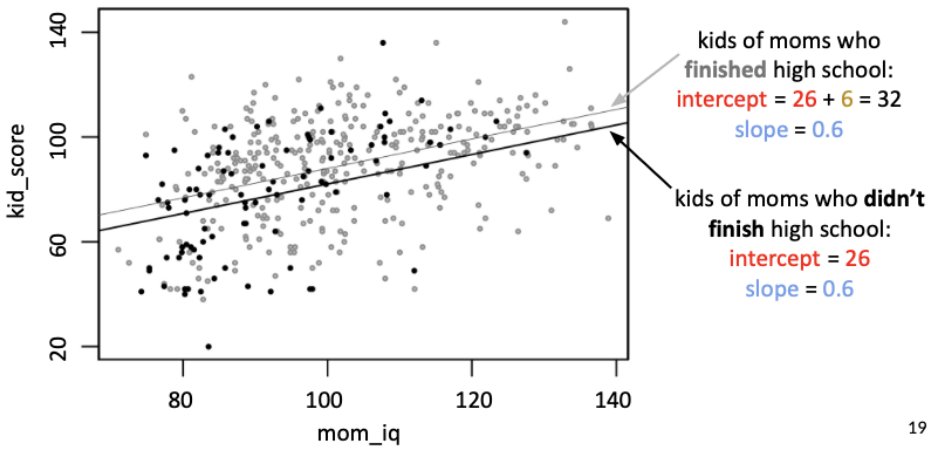


- Intercept β_1: estimated mean outcome for data points i with $X_i = 0$
- Slope β_2: difference in estimated mean outcomes between data points whose X_i's differ by 1
- Why "estimated"? Because sometimes linear regression won't fit our data correctly...
- NB: for binary predictor, we got "exact"instead of "estimated"

Example with multiple predictors

- $(X_{i1} = 1 = \text{constant})$
- $X_{i2} = \texttt{mom_hs} =$ "Did mother finish high school?" $\in 0, 1$, where $0 =$ No and $1 =$ Yes
- $X_{i3} = \texttt{kid_score} =$ child's score on cognitive test $\in [0.140]$
- $y_i = \texttt{kid_score} =$ child's score on cognitive test $\in [0, 140]$

$$y_i = \beta_1 + \beta_2 X_{i2} + \beta_3 X_{i3} + \epsilon_i$$
$$\texttt{kid_score} = 26 + 6 \cdot \texttt{mom_hs} + 0.6 \cdot \texttt{mom_iq} + \text{error}$$



Example with interaction of predictors

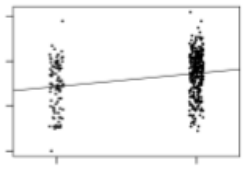
- $X_{i2} = \texttt{mom_hs} =$ "Did mother finish high school?" $\in 0, 1$, where $0 =$ No and $1 =$ Yes
- $X_{i3} = \texttt{mom_iq} =$ mother's IQ score $\in [70, 140]$
- $y_i = \texttt{kid_score} =$ child's score on cognitive test $\in [0.140]$

$$y_i = \beta_1 + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i2} X_{i3} + \epsilon_i$$
$$\texttt{kid_score} = -11 + 51 \cdot \texttt{mom_hs} + 1.1 \cdot \texttt{mom_iq} - 0.5 \cdot \texttt{mom_hs} \cdot \texttt{mom_iq} + \text{error}$$

So why not just compute the two means separately and then compare them?

	Mom drives Mercedes	Mom doesn't drive Mercedes
Mom finished high school	avg kid_score 90	avg kid_score 90
Mom didn't finish high school	avg kid_score 78	avg kid_score 78

	Mom drives Mercedes	Mom doesn't drive Mercedes
Mom finished high school	990 women	10 women
Mom didn't finish high school	10 women	990 women

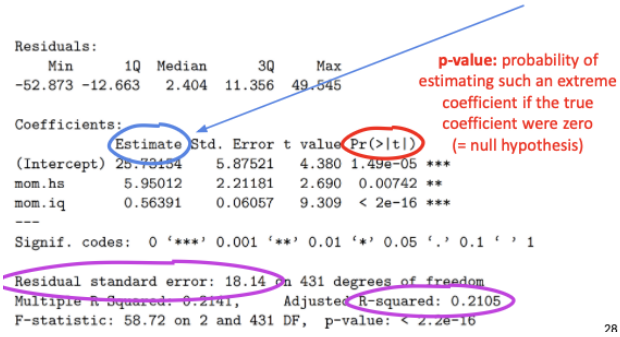


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- Mean `kid_score` for Mercedes drives: $\mathbf{0.99 \cdot 90 + 0.01 \cdot 78 \approx 90}$
- Mean `kid_score` for non-Mercedes drives: $\mathbf{0.01 \cdot 90 + 0.99 \cdot 78 \approx 78}$
- But really driving Mercedes makes no difference (for fixed high-school predictor)!
- Root of evil: **correlation** between finishing high school and driving Mercedes
- **Regression** to the rescue: $\texttt{kid_score} = 78 + 12 \cdot \texttt{mom_hs} + 0 \cdot \textbf{mercedes} + \textbf{error}$

Quantifying uncertainty

- Statistical software gives you more than just coefficient β:



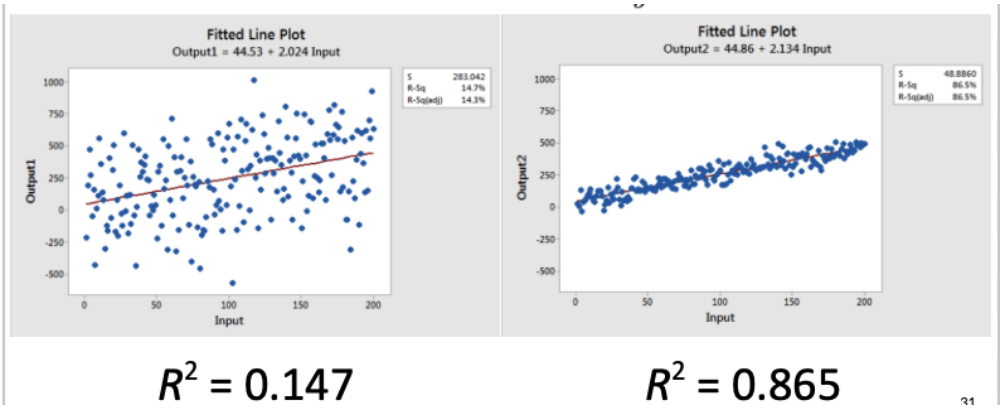
Residuals and R^2

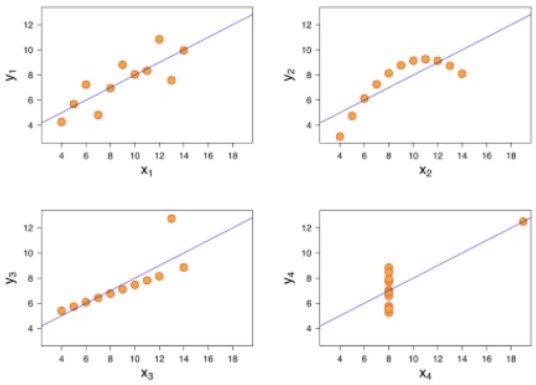
- Residual for data point i: estimation error on data point i:

$$r_i = y_i - X_i\hat{\beta}$$

- Mean of residuals = 0 (total overestimation = total underestimation)
- Variance of residuals
  = avg squared distance of predicted value from observed value
  = "unexplained variance"
- Fraction of variance explained by the model:

$$R^2 = 1 - \hat{\sigma^2}/s_y^2, \text{ variance of outcomes y}$$



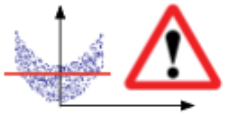


Assumptions made in regression modeling

1. Validity:
   - a) Outcome measure should accurately reflect the phenomenon of interest
   - b) Model should include all relevant predictors
   - c) Model should generalize to cases to which it will be applied
2. Linearity:

$$y_i = \beta_i X_{i1} + ... + \beta_k X_{ik} + \epsilon_i, \text{for i} = 1, ..., \text{n.}$$

But very flexible: we require linearity in predictors (not necessarily in raw inputs); predictors can be arbitrary functions of raw inputs, e.g.,



   - logarithms, polynomials, reciprocals, ...
   - interactions (i.e., products) of multiple inputs
   - discretization of raw inputs, coded as indicator variables

3. Independence of errors: no interaction between data points
4. Equal variance of errors
5. Normality (Gaussianity) of errors

Note: (4) and (5) are less important in practice

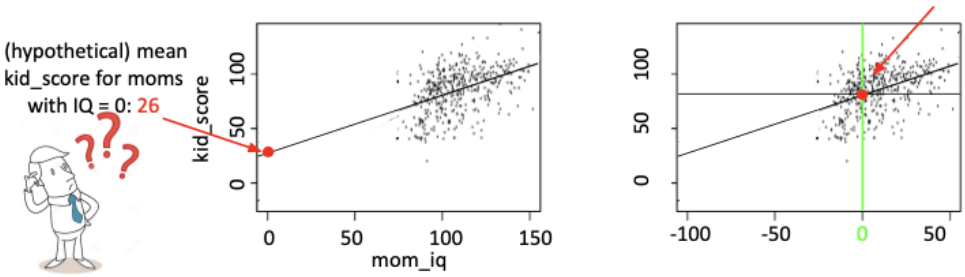
Transformations of predictors and outcomes

Transformations of predictors

- When we apply **linear transformations** to predictors, the model remains "equally good"
  - The fitted coefficients may change, but predicted outcomes and model fit (R^2) won't change

Mean-centering of predictors

- Compute the mean value of a predictor over all data points, and substract it from each value of that predictor:
  $X_{ij} \Leftarrow X_{ij} - mean(X_{1j}, ..., X_{nj})$
- $\Rightarrow$ the predictor X_{ij} now has mean 0



mean `kid_score` for moms with mean IQ: 86

After mean-centering of predictors, you have a convenient interpretation of coefficients β_j of main predictors (i.e., non-interaction predictors):

- j = 1 (i.e., intercept):
  - Estimated mean outcome when each predictor has its mean value
- j > 1
  - Model w/o interactions: estimated mean increase in outcome y for each unit increase in X_{ij}
  - Model with interactions: estimated mean increase in outcome y for each unit increase in X_{ij} **when each other predictor has its mean value**

Standardization via z-scores

- First mean-center all predictors, then divide them by their standard deviations:

$$X_{ij} \Leftarrow [X_{ij} - mean(X_{1j}, ..., X_{nj})]/sd(X_{1j}, ..., X_{nj})$$

- Resulting values are called "**z-scores**"
- All predictors now have the same units:
  distance (in terms of standard deviations) from the mean

- This lets us compare coefficients for predictors with previously incomparable units of measurement, e.g., IQ score vs. earnings in Swiss francs vs. height in centimeters

Logarithmic outcomes

- Pratical: makes sense if the outcome y follows a heavy-tailed distribution (power-law)
- Only works for non-negative outcomes
- Theoretical: turns an additive model into a multiplicative model:

$$\log y_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + ... + \epsilon_i$$

Exponentiating both sides yields

$$\begin{aligned} y_i &= e^{b_0 + b_1 X_{i1} + b_2 X_{i2} + ... + \epsilon_i} \\ &= B_0 \cdot B_1^{X_{i1}} \cdot B_2^{X_{i2}} \cdots E_i \end{aligned}$$

interpreting coefficients

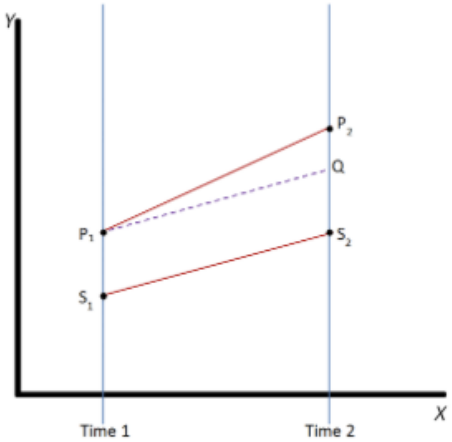
- An additive increase of 1 in predictor $X_i 1$ is associated with a multiplicative increase of $B_1 := exp(b_1)$ in the outcome
- If $b_1 \approx 0$, we can immediately interpret b_1 (without needing to exponentiate it first to get B_1!) as the relative increase in outcomes, since $exp(b_1) \approx 1 + b_1$
- E.g., $b_1 = 0.05 \Rightarrow B_1 = \exp(b_1) \approx 1.05$
  $\Rightarrow$ "+1 in predictor X_{-1}"is associated with "+5% in outcome"

Going beyond linear regression for comparing means

Generalized linear models

- Logistic regression: binary outcomes
- Poisson regression: non-negative integer outcomes (e.g., counts)

- Two groups: P, S
- At time 2, group P receives a **treatment**, group S doesn't
- Question: Did the treatment have an **effect?** If so, how large was it?
- P and S don't start out the same at time 1
- There is a temporal "baseline effect" even w/o treatment



- Elegant linear model with binary predictors:

$$y_{it} = a + b \cdot treated_i + c \cdot time2_t + d \cdot (treated_i \cdot time2_t) + error_i$$

- d = treatment effect
- All of this with one single regression!
- You get quantification of uncertainty (significance) for free!

